

Translating Human-Centred Artificial Intelligence for Clinical Decision Support Systems into Practice: A Medical Retina Case Study.

Josie Carmichael

UCL Interaction Centre (UCLIC), London, United Kingdom and NIHR Biomedical Research Centre at Moorfields Eye Hospital NHS Foundation Trust and UCL., United Kingdom
josie.carmichael.20@ucl.ac.uk

ABSTRACT

Artificial intelligence has potential to enhance healthcare outcomes through applications such as early diagnosis and enhanced treatment planning but can only deliver such potential if it is integrated into clinical practice. Using eyecare as a case study, my research explores optometrists' requirements for an AI clinical decision support system (AI-CDSS) and addresses the 'gap' between AI design motivated by research success vs clinical application. For example, findings from an interview study with 20 optometrists highlighted that clinicians' risk-averse tendencies can significantly affect their interpretation of AI outputs when making clinical decisions. The way in which outputs are presented should neither encourage sub-optimal risk-averse behaviours nor convey misleading information. In the latter stages of my Ph.D., I aim to further investigate how the level of risk associated with clinical decisions can affect interpretations of AI support, as well as testing available methods for improving the interpretability of AI outputs.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools; • **Applied computing** → Life and medical sciences; Health care information systems.

KEYWORDS

Clinical decision support, artificial intelligence, medical imaging, human-AI interaction, healthcare, interpretability

ACM Reference Format:

Josie Carmichael. 2024. Translating Human-Centred Artificial Intelligence for Clinical Decision Support Systems into Practice: A Medical Retina Case Study.. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3638182>

1 INTRODUCTION

Artificial Intelligence (AI) driven technologies are rapidly developing within the field of medicine and have the potential to benefit healthcare through applications such as early and accurate diagnosis [4], enhanced treatment planning [17] and reducing clinician

workload. However, despite many systems demonstrating impressive levels of accuracy for set clinical tasks, most research has been predicated on the comparison between the diagnostic accuracy of AI and humans. This competitive view is unrealistic and research efforts in HCI have therefore emphasized the importance of a user-centric approach, so that AI applications can empower individuals through human-AI collaboration [24]. Machine learning (ML) based systems for healthcare are uniquely difficult to design, develop, and deploy [22], with their uptake influenced by a complex interplay of factors encompassing data quality, regulation, ethics and perhaps most importantly stakeholder acceptance [1]. Understanding the trust dynamics between AI and humans is also crucial to encourage stakeholder acceptance [16] and designing systems that promote appropriate levels of trust in AI systems is a key enabler for deployment in the real world [18]. Due to the ubiquitous use of ophthalmic imaging, its interpretation provides an exemplary instance of an application where an AI-clinical decision support system (AI-CDSS) may prove highly beneficial, through assisting clinicians with the detection and monitoring of ocular diseases [8]. Thus, it is important to understand clinicians' AI support needs. Given the increasing availability of AI solutions, the usability and usefulness of such systems for real-world scenarios requires exploration and may help determine barriers and enablers to implementation. My clinical background as an optometrist, and research foundations in HCI, allow me to bring a unique interdisciplinary approach of combining perspectives to tackle this research gap.

2 BACKGROUND AND RELATED WORK

2.1 Optical coherence tomography (OCT) imaging and possible AI solutions

Recently, there has been an increase in the use of optical coherence tomography (OCT) imaging in eyecare due to its ability to produce a cross-sectional, high-resolution view of the retina. This has led to its widespread adoption in both hospital eye services (HES), and more recently in primary eye care [14]. Improved accessibility to OCT imaging brings about promising advantages by providing supplementary diagnostic information and facilitating the early detection of diseases [10]. However, many optometrists have limited experience in interpreting OCT imaging. As most referrals into the hospital eye service (HES) originate from optometric examinations, the broader integration of OCT imaging within primary care introduces the potential concern of inaccurate interpretations and subsequent unnecessary referrals [11]. Exciting developments in ophthalmic care have been made in AI for OCT interpretation

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3638182>

and analysis. For example, systems have been designed to differentiate various retinal pathologies [8, 13]. Should systems like this be integrated, they could assist optometrists in making more precise decisions regarding patient management. However, despite encouraging viewpoints about the potential role of AI in primary eye care [9], there remains limited understanding of the practical ways in which clinicians might interact with AI in the context of their clinical decision-making.

2.2 Human-AI collaboration in a clinical context

In recent years research has acknowledged that evaluating the usability of an AI-CDSS is more intricate than comparing clinicians' diagnostic performance with and without the inclusion of AI. One study [20] examined whether human-computer collaboration is influenced by the way that AI outputs are presented to humans in skin disorder classification support. They reported significantly improved diagnostic accuracy over that of either AI or physicians alone when AI-based multiclass probabilities were displayed. Despite AI systems demonstrating significant potential for use in a range of medical applications, context-specific challenges often limit the real-world use of these systems. Researchers commonly encounter hurdles in seamlessly integrating AI into clinical workflows, leading to studies and assessments primarily conducted in controlled laboratory settings [23]. For example, Bach et al. [2] underscored that certain principles derived from the bias mitigation literature may not align with the practical challenges faced by clinicians. Wang et al. [21] also evaluated their AI system for diagnostic recommendations in a real-world environment and highlighted several conflicts emerging between the design of the AI-CDSS system and the intricacies of the clinical context. In an attempt to address the mismatches between lab-based and real-world use, Cai et al. [3] explored the differences between individual needs of pathologists when searching for similar images retrieved using a deep learning content-based image retrieval system [3]. They then used these findings to develop human-centered refinement tools to allow users to input to the image retrieval system by communicating what types of similarity are most important to them.

2.3 Interpretability of AI outputs

For clinicians to work effectively with AI, it is important to consider the most appropriate method of displaying outputs to users for AI predictions to both be interpretable and for the system to demonstrate appropriate transparency. Many AI algorithms can be considered 'black box' in nature. This means that they may perform well in terms of accuracy, but understanding the underlying processes used to achieve results can be difficult, even for experts. This lack of transparency may hamper the adoption of AI; as a result, some studies have tried to include techniques to improve the transparency of data-driven AI for pathology, for example by producing "saliency maps" (or "heat-maps"). Although saliency maps have demonstrated good potential, limited research has been carried out in the form of user studies. In ophthalmology research, another potential method has been developed to increase AI transparency for interpreting OCT scans [8]. This method divides the analysis of the OCT scan into two steps to account for inter-patient variability

in disease presentation. As part of the first step, an AI segmentation network is used to create a detailed tissue-segmentation map (Figure 1).

2.4 My related work

Reflecting on my MRes research at UCL is crucial as it served as a foundational year for my Ph.D. That research investigated how an AI-CDSS affected diagnostic responses from optometrists when assessing ambiguous cases of suspected retinal disease. I aimed to assess whether the level of clinician experience in OCT interpretation and the method of displaying AI outputs (i.e., whether segmentation maps were displayed with diagnostic/management suggestions) were factors that affected diagnostic decisions, when compared to a clinical reference standard diagnosis. The paper based on these findings is currently under review at 'Scientific Reports' [5]. For our purposely selected challenging cases, the AI-CDSS did not enhance optometrists' diagnostic performance irrespective of optometrist experience. Optometrists reported higher levels of trust in the AI-CDSS when segmentation maps, as an intermediate step, were displayed along with diagnostic suggestions. In writing up the project and involving other stakeholders, my supervisors and I have gained an in-depth insight into the AI model and have reflected AI design more broadly. For example, diagnostic algorithms often achieve various levels of accuracy for detecting different conditions in multi-class diagnoses, thus the percentages for diagnostic outputs are not directly comparable across conditions. The findings highlighted that determining an appropriate method of displaying AI outputs to users is more complex than first thought.

3 OBJECTIVES AND GOALS

To implement tools that would be disruptive to optometric practice, there was a need to first determine a clinical need for such a system. Therefore, the first two objectives of my Ph.D. project had a clinical focus:

- **Objective 1:** Evaluate the accuracy of referrals originating from primary care optometric practices and determine this accuracy specifically for retinal conditions.
- **Objective 2:** Evaluate the effectiveness and sustainability of current/previous approaches that have been made to try and reduce the number of false positive referrals seen in the HES.

The above two objectives were addressed during the first year of my Ph.D. project through completing quantitative and mixed-methods systematic reviews. Both reviews are published [6, 7]. The main findings were that optometrists are in fact over-referring patients to the HES and that there is a current lack of an intervention for false positive referrals that appears to 'tick all of the boxes'. This leaves a potential space for newer interventions, such as AI-CDSS. During my Ph.D. project, I will also focus on three specific objectives that aim to contribute towards understanding of how an AI-CDSS could be implemented into primary care practice. These objectives have both a HCI and clinical focus and were studied during the second year of my Ph. D. project:

- **Objective 3:** To determine an appropriate method of displaying outputs from an AI-CDSS to optometrists for them to be clinically useful and interpretable.
- **Objective 4:** To determine an appropriate method of displaying outputs from an AI-CDSS to optometrists for them to not be misleading in relation to AI's predictions.
- **Objective 5:** To explore the relationship between the risk of the AI-supported task and the interpretations of the AI outputs.

Although addressing my objectives using a unique use case, my research connects various aspects of a larger, complex problem of the barriers to successful implementation of AI support, particularly in clinical applications. Unlike most other contexts which measure the success of AI systems by comparing them to humans, my research highlights the complexities of human-AI collaboration in real-world application. My background allows me to combine concepts or methods from different disciplines to tackle these research problems.

4 RESEARCH APPROACH

To work towards addressing objectives 3,4 and 5, I completed an interview study with 20 primary care optometrists who were recruited to include a diverse sample (i.e., different years of experience, practising in independent and multiple practices and across different locations in the UK). All interviews were semi-structured and focused on obtaining information in four separate parts: 1. Experience of OCT interpretation and use in clinical workflows, 2. Information seeking for memorable cases, 3. Assessment of example ambiguous clinical cases and 4. Potential for AI use in primary care (through demonstrations for the same ambiguous cases).

4.1 Example AI-CDSS

AI outputs were presented from one example system for OCT interpretation (Figure 1). Full details of the system can be found elsewhere [8]. This system was chosen based on its availability, its ability to detect several retinal pathologies, its presentation of a **segmentation map** which may also improve the interpretability of the system for users and its displaying of three separate outputs: segmentation maps, diagnostic suggestions and management suggestions. In summary, the example system uses two separate sets of algorithms. The first set of algorithms uses the OCT scan as an input and segments it into identified features of anatomy and pathology. These segmentations are superimposed as color-coded overlays atop the OCT scan. Subsequently, the outcomes featured in the segmentation map serve as inputs for a second set of algorithms, responsible for detecting various potential diagnoses. It is important to note that multiple diagnoses can potentially co-exist. For this study, we opted to present each diagnosis result as either 'detected' or not (Figure 1). This simplified presentation approach was chosen based on key considerations associated with the architecture of the diagnostic algorithm.

4.2 Data analysis

A subset of the data warranted a basic quantitative analysis. The intention behind this quantitative approach was to first obtain a streamlined summary of the engagement with AI. The primary

methodology employed for qualitative data analysis encompassed an inductive approach, characterized by open and iterative coding and determining dominant themes.

5 SUMMARY OF FINDINGS

A more detailed summary of the findings has been reported in a research paper submitted to the CHI 2024. In this report I summarise the most novel findings and discussion points that will influence the latter stages of my Ph.D. project.

5.1 AI's influence on diagnostic decisions

In 9/19 (47%) of instances where optometrists viewed an AI's diagnosis that was contradictory (but correct compared to a reference standard) to their initial assessment, they changed their original diagnosis. Although a simulated scenario, this indicates an influence of the AI on some optometrists' interpretations of OCT imaging. The decision switches in our study appeared to be linked to optometrists being unsure in their assessment. These switches are important to consider as most participants stated that they view OCT imaging at the beginning of the eye examination. If presented with AI diagnostic suggestions based on OCT findings at this stage, this may create anchoring bias [2] towards that output and shape the eye examination and decisions that follow.

5.2 Intended use of AI

The results highlighted that the perceived usefulness of and interactions with an AI system can be significantly influenced by its intended purpose. For our example use case we considered whether the AI's purpose was solely to aid clinicians in making sense of OCT findings or to provide support at the stage of making more complex patient management decisions. The interpretation of specific OCT features, such as determining whether an area in the retina is swelling or not, can be made in isolation. Segmentation maps serve as foundational elements aiding optometrists in their patient assessments. These offer the advantage of identifying such features independently from external influences and can focus clinicians' attention on pathological features without making recommendations. The second way in which AI could support optometrists is to suggest an overall diagnosis and management decision. However, of the three outputs, management suggestions were considered the least useful, as offering AI management suggestions solely based on the OCT was not regarded as satisfactory holistic support.

5.3 Research vs practice

In the context of AI-assisted decision-making, behaviour is commonly characterized as encompassing a restricted set of choices (such as a diagnosis) [19]. However, decisions regarding patient diagnosis and particularly management entails the consideration of several interconnected factors. The development of a system that takes these factors into account could potentially enhance the diagnostic precision of an algorithm; but it may be impossible for current AI technology to encompass the complexity of clinicians' holistic interactions with patients. For example, patient-centered questions such as 'does he have a high-pressure job?' may hold varying importance to different clinicians. The 'gap' between AI designed for research vs clinical application is also apparent when

systems are optimized for specific tasks/conditions. While research serves as a foundational step in AI development, the intention for a system plays a significant role in how it is designed and assessed. In research, the primary intention is often to demonstrate the feasibility of a new AI system, and the easiest way to achieve this is to compare to humans as a benchmark for high-performance. However, systems are commonly not designed to adequately meet the context-specific challenges of real-world use.

5.4 The influence of risk-taking

One common theme from participants was a concern about risk taking. For example, participants talked about erring on the side of caution due to the clinical risks and this behaviour was also witnessed during the assessment of example cases. These risk-adverse tendencies could elucidate why optometrists were positive about the illustrated approach of displaying diagnostic outputs (detected vs. not detected) for our selected clinical cases and why they did not express apprehension about the absence of confidence values accompanying the diagnostic outputs. This should be considered

when deliberating on the way AI recommendations should be presented to medical professionals as the way in which the decision and its level of associated risk is ‘framed’ is important to consider, especially in the context of risk and uncertainty [15]. This contrasts non-clinical applications of AI support such as bird classification, where a frequently expressed need from users was for the AI to display its confidence to better determine when to trust the AI’s output [12]. In clinical applications there needs to be a balance between outputs not being misleading but also not encouraging suboptimal risk-averse behaviours.

5.5 Conclusions

My research contributes to the growing HCI literature in translating AI systems from research to practice through investigating real-world user needs. In the example use case, segmentation maps helped optometrists to interpret the OCT using simplification and thus they viewed the segmentations as a positive addition which can assess the OCT in isolation. Patient management decisions are influenced by risk-adverse tendencies of clinicians and can cloud their probabilistic reasoning when viewing AI outputs. This is

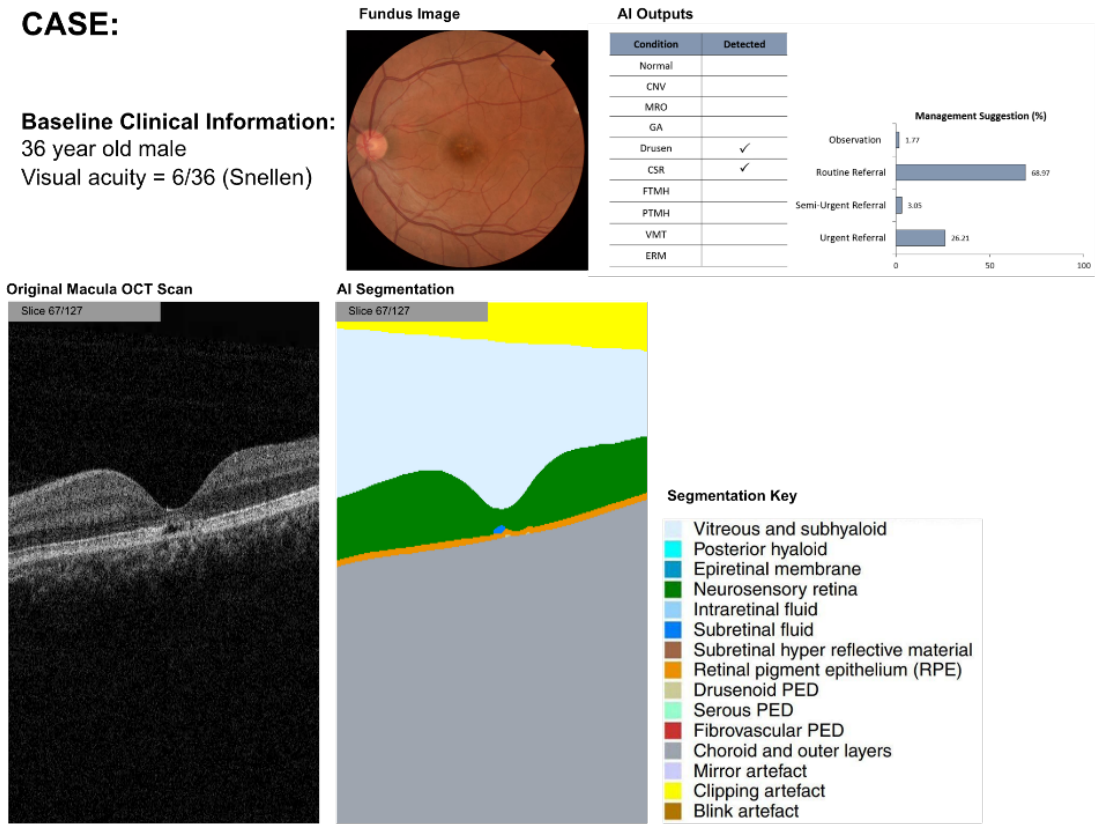


Figure 1: Information demonstrated to participants during stage 5 of the interview study. ‘AI Segmentation’ displays segmentation overlays produced by an AI system which highlights color-coded anatomical and pathological OCT features. ‘AI Outputs’ displays the AI’s diagnostic and referral suggestions using multi-class probabilities. These probabilities are displayed as a percentage out of 100%

important to consider, as outputs should neither encourage suboptimal risk-averse behaviours nor convey misleading information. It may be impossible for AI to encompass the complexity of clinical decisions and thus the answer may be to more clearly define the isolated task that AI is designed to support.

6 NEXT STEPS

The interviews highlighted important barriers and facilitators to the implementation of AI-CDSS in eyecare and raised important discussion points as described above. I next aim to determine the most appropriate methods for AI output interpretability in my use case, through demonstrating different available options. A co-design workshop has been considered as a suitable method to explore insights from clinicians. The advantages of using workshops are the ability to gain input from potential future users of the AI system in a relatively short period of time, and the richness and quality of the output that derives from participants' discussion and idea stimulation. After determining the most appropriate method for optimizing the interpretability of AI outputs for OCT imaging, I then aim to carry out a more focused study of the different options for displaying AI diagnostic outputs and how level of risk associated with a clinical management decision may affect clinicians' interpretations of, and agreement with outputs. This would be carried out as a vignette study. I believe this focus is novel and would make an important contribution to the HCI literature.

7 LONG TERM GOALS

I aim to present the results of my Ph.D. dissertation using three distinct sections of my dissertation. The first will discuss evidence of the need for AI-CDSS in my use case. The second will discuss the exploratory approach used in the interview study to identify several key barriers to the implementation of an AI-CDSS. The third (detailed above) will have a strong HCI focus, during which I aim to develop my design skills. The CHI doctoral consortium will therefore be a hugely beneficial step. Post-Ph.D., my longer-term vision would be to continue researching human-AI collaboration in healthcare using the skills learned during the Ph.D. The ideal goal would be to obtain funding to explore more AI design-related factors and to use these findings to produce a functional system, before carrying out a trial of this system in clinical practice.

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